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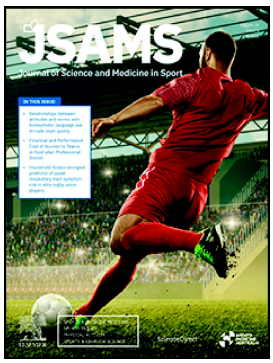

Estimating the cost of sports injuries: a scoping review

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Estimating the cost of sports injuries: a scoping review

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Estimating the cost of sports injuries: a scoping review

ABSTRACT

Objective: To provide an overview of the methods used to estimate the cost of sports-related injury published to date, and to highlight considerations and opportunities for future research.

Design: Scoping review.

Method: Scopus, MEDLINE and CINHAL were searched from 1st January 2000 to 1st January 2023. Studies were screened by two independent reviewers and were eligible if they reported on a cost analysis or cost estimation of sports related injury.

Results: Thirty-one studies fulfilled the inclusion criteria. Twenty-seven studies (87%) were published since 2014. The type of costs included direct healthcare costs (12 studies), indirect costs (10 studies) and a combination of both (9 studies). Twenty-one studies (68%) used a bottom-up costing approach to measure costs of sports injury and estimated direct costs from the service rates or fee schedules of health systems, hospital, insurance companies or national insurance boards. A top-down approach was used in seven studies (23%) to estimate the indirect salary cost of time-loss injuries using data from publicly available resources. Ten studies were from the cost perspective of a sporting organisation (32%). There was a lack of explicit reporting of the costing method used and the perspective of those bearing the costs.

Conclusion Estimating the cost of sports injuries is an emerging area of research, with publications increasing in recent years. However, there remains a lack of methodological guidance to inform or appraise these studies. The expansion of established cost of illness checklists with sport injury explanations to guide future cost of sports injuries studies is recommended.

Keywords: Economics, athletes, medicine, illness, financial burden

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INTRODUCTION

Consequences of sports-related injuries not only impact on athletic performance outcomes[1], but are also associated with a range of financial costs. Healthcare expenses, as well as lost productivity due to absenteeism, can result in considerable financial burden to the athlete, the team, the sports organisation, and the wider community[2]. Identifying and classifying the financial costs associated with sports injury provides opportunities to establish the extent to which sports injuries are an economic problem and for whom, direct prevention priorities related to specific injuries and sports, and provide a means to monitor trends and evaluate the return on investment of implemented injury prevention measures[3].

'Cost-of-illness' (COI) studies are a widely used approach to measure the economic burden of disease in monetary units. Traditional COI studies include both direct healthcare-related costs, as well as indirect costs arising from lost productivity[4]. An understanding of COI can inform health care policy, intervention prioritisation and assist in the efficient allocation of resources [5 6]. COI studies are diverse in design, meaning that inconsistencies in the way studies are conducted and reported can make them difficult to interpret [7 8]. Within the broader medical context, prior studies have called for a standardisation of COI methods to guide the interpretation and critical evaluation of COI studies to better position cost analysis as a useful tool to guide decision making practices [5-7 9 10].

COI research has extended to include sports injuries. However, there are a limited number of cost of sports injury studies in the scientific literature[2 11] and a COI approach to sports related injury research has not been described. A need for high quality, cost of sports injury research has been identified as a priority for future research[3 11 12]. Therefore, this scoping review aims to provide an overview of the terminology and methods used to estimate the cost of sports-related injury in the published literature, to identify common themes and gaps in the evidence, and offer considerations for future cost of sports injury research studies.

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METHODS

This scoping review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Extension for Scoping Reviews (PRISMA-ScR) checklist [13]. A protocol was developed and registered 23rd March 2023 with Open Science Framework (https://doi.org/10.17605/OSF.IO/9YGJF).

Three electronic databases (Scopus, MEDLINE, CINHAL) were searched from 1 January 2000 to 1 January 2023. The period limitation was due to the contemporaneous nature of the topic. The search strategy used the following terms, limited to title and abstract fields, according to three main concepts: (1) financial cost ('Cost and cost analysis' OR 'financ*' OR 'cost*' OR 'economic*' OR 'expense*' OR 'expenditure') AND (2) Injury ('Injur*' OR 'ill*' OR 'poor health' OR 'Health problem*') AND (3) Sports ('Professional sport*' OR 'elite sport*' OR 'Olympic sport' OR 'professional athlet*' OR 'elite athlet*' OR 'Olympic athlet*' OR sport* OR athlet*). The search strategy was modified according to the specifications of each database. There were no language limitations set and handsearching was used to identify any further potentially eligible studies. For this review, only studies that performed cost analyses or cost estimations of sports-related injuries were included. Studies investigating the outcomes, or cost effectiveness, of injury prevention or management programs were excluded. A detailed eligibility criteria is provided in the online supplementary file. Database results were exported to Covidence online (www.covidence.org) for screening and data charting. Titles and abstracts retrieved from the databases, as well as those found from handsearching, were independently screened for eligibility by two reviewers (MT and TG) using the pre-specified eligibility criteria. Articles assessed as eligible according to title and abstract were then independently screened by full-text review by the same two reviewers, with any discrepancies resolved through consensus.

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A data charting form was created and tested, with the content agreed on by all authors prior to commencing data extraction. The extracted data was exported by CSV and Microsoft Excel was used to categorize studies by study design; type of sport; level of participation; injury definition; injury diagnosis; injury diagnosis code; type of cost: direct and indirect costs; cost approach: bottom-up approach, top-down approach, econometric approach; cost perspective; and year of currency. Key definitions are provided in Table 1. Where data items were not explicitly reported in the included articles, where possible they were extracted based on interpretation. Data were synthesized using narrative and descriptive summaries. To categorise cost of sports injury studies by injury definition, three constructs of sports injury and illness definitions were used: 'athlete self-report', 'medical attention' and 'sports incapacity' (time-loss) as described by Bahr et al[14]. COI literature[6 7] was used to assist in identifying and categorising the costing approaches and cost perspectives when these were not stated or reported within the included cost of sports injury studies' methods.

RESULTS

The search strategy revealed 8,193 studies, of which 810 were excluded as duplicates (Figure 1). Of 7,383 titles and abstracts screened, 70 articles were eligible for full text review. Thirty-nine articles were excluded during full-text screening, leaving 31 studies for inclusion (Table 2). Of the 31 included studies, most had a retrospective data collection design (n=23, 74%)[15-37] and eight studies were prospectively conducted[38-45] (Table 2). The majority (n= 27, 87%) adopted a cohort study design, there were two cross sectional studies[23 24] and two descriptive epidemiological designs[25 31]. Study settings included elite or professional sports league observations (35%)[15-18 21 23 24 30 34 42 43], hospital admissions of sports related injuries (19%)[25 26 31 32 37 38], elite and amateur sports injury-claim settings (23%)[19 20 22 28 29 33 35], private clinic presentations (6%)[27 36], sports tournaments (3%)[45] and individual athlete programs (13%)[39-41 44]. The year of publication ranged from 2005 to 2022. Twenty-seven studies (87%)

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were published in the nine years since 2014[6 15-18 20 21 23-32 34 36-45], of which 12 (39%) were published since 2020[15 21 23 26 27 29 31 34 36 38 41 42].

The time horizon of the included studies varied from less than one year (2 of 31 studies, 6%)[39 44], one year (8 of 31 studies, 26%)[19 29 31 35 38 40 41 43], two to five years (5 of 31 studies, 16%)[23 24 28 37 45], six to 10 years (11 of 31 studies, 35%)[16 20-22 25 27 30 32 33 36 42] and 11 to 20 years (5 of 31studies,16%)[15 17 18 26 34]. The two studies of less than one year analysed the injury costs of running training programs[39 44]. The longer-term studies that were 11 to 20 years included the costs of long-term outcomes of sports injury. For example, Ackerman et al[26] quantified the likelihood and costs of hip replacement surgery after sports injury over a 15-year period.

Eleven studies (35%) sourced injury and/or financial data from publicly available resources[15-18 20 21 23 24 30 34 42]. For example, nine of these studies involved professional sports settings including the Australian Football League (AFL)[30], Major League Baseball (MLB)[15 21], National Hockey League (NHL)[16 18 24], English Premier League (EPL)[23], National Basketball Association (NBA)[34] and National Football League (NFL)[17], which estimated the indirect salary cost of injury using data obtained from websites or the sporting organisations annual report. Twelve studies (39%), including hospital[25 26 31 32 37 38], injury insurance claim[19 20 22 28 29 33 35] and private clinic[27 36] study settings, used internal administration or organisation data. For example, Robinson et al[31] used internal electronic health records to identify sports related hand and wrist injuries and used hospital billing records to extract the costs of all resources used from the injury onset to its disappearance including inpatient costs and outpatient costs. Eight (26%) studies conducted questionnaires or surveys[29 35 39-41 43-45]. For example, Traneaus et al[43] after identifying injuries to Floorball athletes, used questionnaires to link the costs of the resources used to treat the injury.

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The level of sports participation included 13 professional or elite[15-18 20-24 30 34 42 43], one sub-elite[45], five recreational or amateur[29 39-41 44] and 12 studies that did not specify the level[19 25-28 31-33 35-38]. The type of sports studied included running[39-41 44], ice hockey[16 18 24], football/soccer[23 29 46], rugby union[28 45], horse racing or equestrian[22 37], baseball[15 21], floorball[43], rugby league[33], gridiron[17], Gaelic games[20], basketball[47] and Australian football[30] (Table 2). Five studies investigated multiple sports [19 27 31 36 48] and four studies identified sports-related injuries, however the type of sport was not specified[25 26 32 38].

Fifteen studies (48%) used a sports incapacity (time-loss) definition [15-18 21-24 30 34 39 42 43 45], 14 (45%) used the medical attention definition [19 20 25-29 31-33 35-38 41], and two studies (6%) used an athlete self-report definition[40 44]. Diagnostic coding systems were reported by only 10 studies (32%), where one study used the International Classification for Disease version nine (ICD-9)[27], six studies used the International Classification for Disease version ten (ICD-10)[25 31 32 36 38 43], and three studies used the Orchard Sports Injury Classification System (OSICS) [30 40 44]. Nine studies (29%) identified a single specific diagnosis for cost analysis without reference to a coding or classification system[15-18 21 26 30 34], nine (29%) included multiple body areas injured and broadly classified the structure injured[19 20 22 24 28 29 33 37 41 45], and three (10%) did not clearly specify the injury diagnosis[23 39 42].

The injuries investigated according to body area were wrist or hand[31], thigh[30], shoulder[15], knee[34-36], hip[26], elbow[21], ankle[27], head[16-18], multiple areas[19 20 22 24 25 28 29 32 33 38 40 41 43 45] or not specified[23 39 42] (Table 2). Specific injury diagnoses were detailed in several studies and included hamstring strains[30], shoulder superior labrum anterior-posterior (SLAP) tears[15], knee anterior cruciate ligament (ACL) sprains[34], hip arthritis[26], elbow ulna collateral ligament (UCL) sprain[21], ankle sprains[27] and concussion[16-18].

Twelve studies investigated direct costs of sports injury[25-27 31-33 35-38 43 45], 10 studies examined indirect costs[15-18 21 23 24 30 34 42] and nine studies incorporated both direct and

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indirect costs of sports injuries[19 20 22 28 29 39-41 44] (Table 3). There was a variation in the type of direct healthcare evaluated, with some studies accounting for more comprehensive resource use than others. For example, the majority of hospital setting studies included hospital inpatient services and outpatient services including physician services, radiology, pathology, pharmacy and ancillary services such as allied health, nurses and ambulance[25 26 31 32 37 38]. Whereas two private clinic setting studies analysed the costs of athletic trainer services[27 36]. Of the studies in sporting organisation settings, the majority (9 of 10 studies, 90%) examined productivity loss only.

Most studies (n=21, 68%) used a bottom-up approach[25-29 31-33 36-39 41 44 45] to measure direct and indirect costs of sports injury (Table 2). In the studies where a bottom-up approach was used to measure direct costs, the unit costs of direct healthcare resources were estimated from the service rates or fee schedules of health systems, hospital, insurance companies or national insurance boards[25-29 31-33 36-39 41 44 45]. The hospital setting studies also used internal administration or organisation data to identify the average daily cost per Diagnosis Related Groups (DRG), episode of care type and length of stay to estimate service costs[31 32 35 37]. A number of studies utilised actual unit costs of healthcare as identified by the athlete participants in the study[19 40 43] and insurance claims[20].

The costing approach used to measure indirect costs in the included studies varied. For example, the studies that estimated indirect salary costs using a top-down approach divided annual salary expenditure by the number of players per team and the number of matches or days per season to work out an estimated weekly salary per player, which was then multiplied by the number of matches or days missed per injured player to estimate salary cost of injury[15 18 21 24 30 34 42]. Conversely, other studies applied a bottom-up approach to measure opportunity costs of absenteeism using estimates of lost productivity[19 20 22 28 29 39-41 44]. The human capital method was used, however only one study explicitly stated its use in their methods[19]. Cumps et al[19] stated the use of the human capital method to determine the

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indirect costs of sports injuries in Flanders, whereby the absenteeism from work was expressed in days of absence and multiplied by the socioeconomic cost inherent to production loss. Two studies used an econometric approach[16 17] where the difference in cost of an injured group was compared to the non-injured group.

Ten cost of sports injury studies were from the perspective of a sporting organisation (10 of 31, 32%), six were from the perspective of a third-party payer (19%), six were societal (19%), six from a health system or hospital perspective (19%), and three were athlete or athlete families (10%). These results were interpreted based on the types of costs included as only one of the included studies[31] specifically stated a cost perspective.

DISCUSSION

The absence of standardised methodological approaches specific to cost of sports related injury is apparent with a moderate number of heterogenous publications identified in this scoping review. The topic of sport-related injury cost appears to be an emerging area of interest, with most articles (27 of 31 studies, 87%) published in the last nine years and 12 studies (39%) published since 2020. The review identified a range of inconsistent methods utilised by cost of sports related injury studies including how sports injuries are defined and classified, the types of costs included, the approaches used to measure these costs and the perspectives from which the costs are borne. Improving research approaches will assist in advancing cost of sport related injury studies.

Identifying sports injuries

Clearly defining what constitutes a sports injury allows for data to be compared between studies and enhances data collection and research reporting[14]. A range of injury definitions was reported in the cost of sports injury studies, including time-loss and medical attention definitions. One consideration for future cost of sports injury research is that the time-loss injury definition may capture only a proportion of all medical attention injuries[49 50], thus in the

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context of cost of sports injury research may lead to underestimating the true financial cost of a specific injury. Injuries that do not cause time-loss typically require a quantum of healthcare to manage and the direct costs associated with subsequent conditions may be unaccounted for. For example, an injury that does not lead to time-loss from sport such as an Achilles tendinopathy, may require regular physiotherapy visits to assist in managing the injury.

Injury diagnosis was typically not reported or was insufficiently reported in most studies. This results in sports injury being categorised into a general 'undiagnosed category' and potentially reduces cost estimation accuracy. For studies that did specify an injury diagnosis, this was either classified by body area and broadly by an injured structure or incorporated the use of specific diagnosis allocation. Only 10 studies used an injury classification system. Hickey et al[30] for example extracted data from the AFL injury report [50 51] that used the OSICS to group hamstring strain diagnoses to determine the incidence and prevalence of hamstring strain injury. The use of injury classification systems in sports injury surveillance systems may enhance future cost of sports injury research through greater injury specificity. More accurate classifications and group diagnoses can inform databases from which more specific types of injury may be extracted in future financial cost research[14]. Understanding the distribution of injuries relative to diagnoses is an important consideration in generalisability and season-to-season comparisons. An example of this would be a team which has an atypically high number of ACL injuries in one season in which the costing was used. The study may be inappropriately generalised to all seasons or other clubs.

The types of costs estimated

The selection of the type of cost used within the studies was dependent on the available data and the study objectives[7 52 53]. Direct and indirect cost measures independently provide useful information that guide decision making practices[5]. However, compared with direct healthcare costs, direct non-healthcare and indirect costs are more challenging to collect as

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data are often less readily available[9]. Furthermore, methodological issues in costing productivity losses, including issues relating to equity, perspective and valuation methods, has led to ongoing debate as whether and how to include productivity loss in COI studies[54]. These debates may extend to cost of sports injury studies. For example, the cost of a hamstring injury in AFL players will be higher for those players who are on a larger salary. Additionally, the use of friction cost approach may lead to substantially smaller estimates than human capital approach due to the presence of reserve players within a team or squad, making replacement costs negligible. The question of whether and how to include productivity loss will ultimately depend on the study objectives and perspective, and should therefore be considered within the context of a particular study.

The cost of sports injury studies that measured direct healthcare costs varied in their approaches to resource use, with some studies accounting for more comprehensive resource use than others, making direct comparisons between studies difficult. Studies did note this as a limitation. Tranaeus et al[43] for example highlighted that treatments by medical personnel and other support staff that are employed by the team were not measured. Tranaeus et al[43] highlighted that athletes can access various formal and informal healthcare from team staff which makes it uncertain of the true number of treatments an athlete receives. While difficult to measure, researchers should consider capturing all direct resource use items in-order to reduce the risk of underestimating costs of sports injury.

An important observation is that there were international differences in healthcare system coverage across the studies and these influenced the type of healthcare resource use and costs. For example, two New Zealand studies utilised the Accident Compensation Corporation (ACC), a system of universal health coverage to identify the healthcare resource costs of sports injuries[28 33]. By contrast other studies utilised a mix of public and private healthcare or user pay systems. The nature of a countries health systems therefore affects both the availability of healthcare resource use for the injured athlete and the costs where

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some service costs are subsidised, and others are not. It is therefore important to consider the differences in sports study settings including differences in health coverage systems that vary between countries.

Bottom-up versus top-down costing approach

An important decision in costing is whether to use a top-down or bottom-up approach[52]. Benefits of both approaches have been reported[55], however the two approaches capture different types of costs and care should be taken when mixing these methods[52]. The bottom-up approach has been reported to have greater accuracy compared to the top-down approach, as it more comprehensively captures the resources used[52], noting that the decision on costing approach depends on data availability and the study objectives[7 52 53]. However, a bottom-up approach typically requires more granular resource use and expenditure data, compared to the top-down approach that allocates a proportion of a known total expenditure[56]. This is evident in the studies that estimated indirect salary costs of sports injuries in professional sports using a top-down approach[15 18 21 24 30 34 42], which used publicly available total annual salary expenditures to calculate mean weekly salary costs of match-loss injury. The limitation of these studies is that they potentially overestimate or underestimate the true athlete salary costs as individual athletes may be on grossly disparate incomes, or on different contract agreements. In these examples, if there was access to actual player salary from internal administration or organization data, then a bottom-up approach may provide more accurate valuation.

Human capital method for estimating indirect costs

Only one study was identified that explicitly reported the use of the human capital method[19]. While the remaining studies that estimated indirect costs of sports injury did not report their use of this approach, the same method was applied in estimating the value of potentially lost production as a consequence of sports injury. For example, a number of studies used matches missed due to injury, as a proportion of the total number of regular season

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matches, to estimate the salary costs, or lost productivity costs of sports injury[15 18 21 24 30 34 42]. The only study that stated the use of the human capital method[19], also considered its potential limitation in that it fails to account for a replacement worker. This approach has been criticized in the broader COI literature for overvaluing the indirect costs[5 6 57]. The friction cost method has been discussed as a way of preventing further production losses[19], however this method is rarely used in COI research as it requires extensive data to estimate only the losses occurring during the friction period[6]. Despite criticism in the broader literature, the human capital method is adopted by most COI researchers[6] and in the context of sports injury research it should be considered for measuring indirect costs within the known limitations.

Cost perspective

Only one of the included studies explicitly stated a cost perspective[31]. We therefore interpreted the cost perspectives based on the types of costs included. For example, Sleeswijk et al[41] estimated the direct and indirect costs of running related injuries, that included multiple bearers of the costs including the athlete, the employer and the health system, and as such was categorised as a societal perspective study. Donaldson et al[24] estimated the indirect salary costs of missing a game due to injury in the NHL from the perspective of the sports organisation or business. A number of studies were from the perspective of third-party payers, or insurance claims[20 22 28 29 33 35].

The cost perspective of each study provides useful information about the costs to particular groups. Studies conducted from the cost perspective of a sports business for example, provide useful information to guide decision making practices for key stakeholders within those sports businesses. Nonetheless, by focussing on salary-related costs alone, the full impact on healthcare resource use is likely underestimated. Health economists typically recommend a broader societal perspective that accounts for direct and indirect costs for all members of a given society[6 7], minimising the potential biases of narrower views[7]. A societal

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cost perspective is not always possible as it often requires extensive data and may be difficult to undertake when there are low injury cases[6]. In the context of sports injuries this could pose challenges due to the diversity and multiple levels of sport participation. There was a lack of explicit reference to costing perspective in the sports injury literature and this should be addressed in future research.

Checklists and methodological guidelines exist to aid the appraisal of COI research. A recent consensus-based checklist by Schnitzler et al[10] was published in 2023 to standardise the critical appraisal of COI studies. Schnitzler et al[10] provided supporting guidance statements to a checklist across three domains: study characteristics, methodology and cost analysis, and results and reporting. Among these items were perspective, epidemiological approach, costing approach, data collection and resources including cost types. This checklist is recent and was developed from a six-step process including a scoping review, comparisons of existing COI checklists and expert health economist interviews. We recommend that this checklist or an extension of it be developed and applied to cost of sports injury research. Additional considerations may be required for an extension to cost of sports injury studies. For example, the epidemiological approach item may require an additional statement regarding the identification of a sports injury and what constitutes a sports injury. A COI checklist applied to cost of sports injury studies would address the heterogeneity highlighted in this scoping review by promoting consistency and allow for improved comparability of future cost of sports injury studies. Furthermore, collaboration with a health economist is recommended to assist sports medicine researchers in this emerging area.

Limitations

The results for cost approaches and cost perspectives across the cost of sports injury studies should be interpreted with a degree of caution as most studies did not state these explicitly and as such were determined by the authors of the review. This review only considered sports injuries and therefore did not include sports-related illness. The inclusion

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of illnesses should be considered where possible in future cost of injury research. The search strategy was restricted from 1 January 2000, which may have resulted in a small number of studies being excluded. However, this decision was made due to the contemporaneous nature of this topic, which is evident by the recency of the majority of the included studies. Furthermore, the risk of study bias and reporting quality of each of the included studies was not reviewed, as these were not objectives of this scoping review. As the methodology advances in sport-related cost of injury studies, further guidance may be required to inform statistical analysis methods and reporting.

CONCLUSION

Despite the apparent increase in research investigating the cost of sports injuries in recent years, there remains a lack of methodological guidance for these studies. We recommend that future cost of sports injury studies consider the use of and provide clear statements regarding injury or illness identification, the type of costs investigated, the choice of costing approach and cost perspective. We recommend that a consensus-based checklist be applied or expanded to cost of sports injury studies to help guide and appraise future research. This will reduce methodological heterogeneity and improve comparability across cost of sports injury studies. Further quality cost of sports injury research will benefit various stakeholders in the sporting sector, assisting decision making practices including the prioritisation of prevention programs, targeting the allocation of healthcare resources, and informing policy.

PRACTICAL IMPLICATIONS

- This review summarises the methods of cost of sports injury literature and highlights the types of expenses associated with sports-related injuries.

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- Cost of sports injury studies should consider the use of cost-of illness (COI) checklist and provide clear statements regarding injury or illness identification, the type of costs investigated, the choice of costing approach and cost perspective.
- Further cost of sports injury research will benefit various stakeholders in the sporting sector, assisting decision making practices including the prioritisation of prevention programs, targeting the allocation of healthcare resources, and informing policy.

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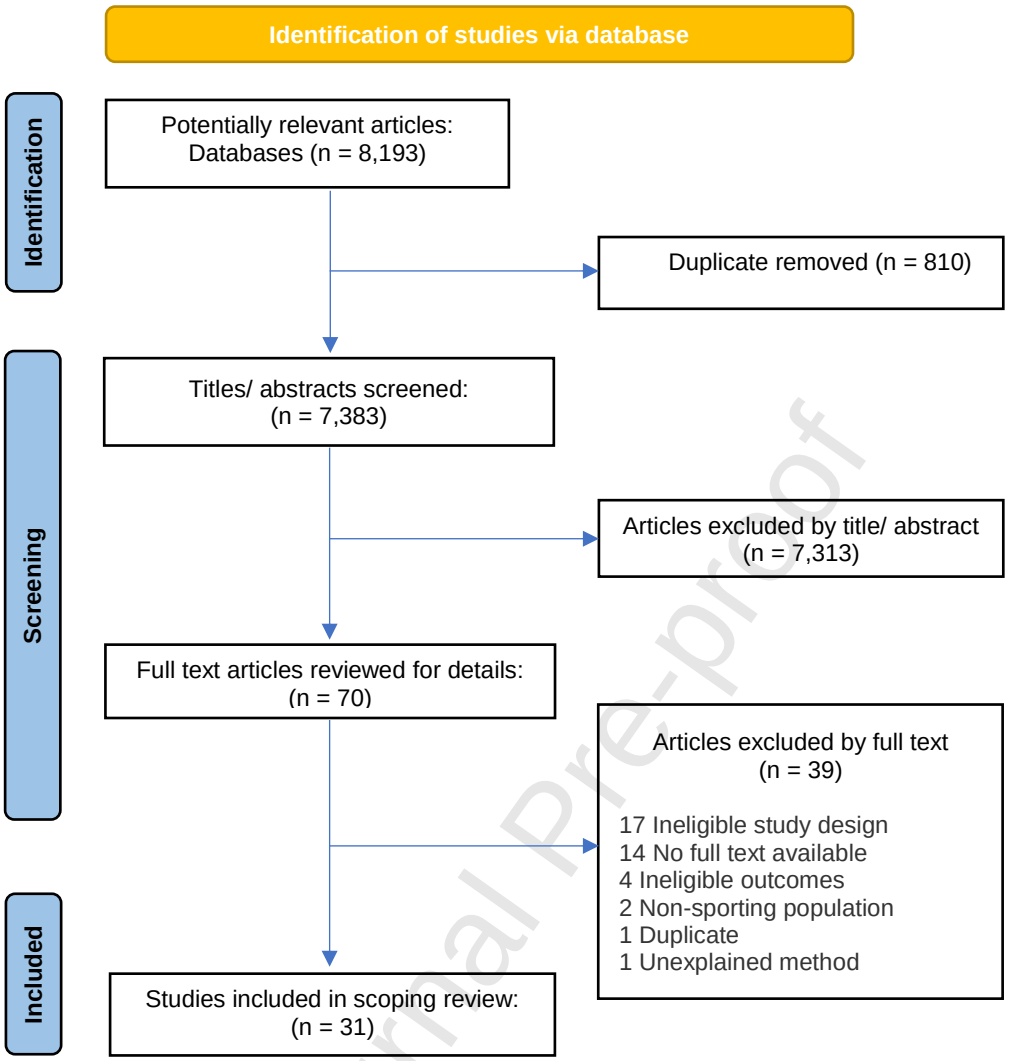

Figure 1: Flow chart of search following the Preferred Reporting for Systematic Reviews and Meta-Analysis guidelines.

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Table 1 Key definitions

Data item	Definition
Direct costs	The cost of the resources used to diagnose, treat and rehabilitate injury or illness[58]. Direct cost resources may be further categorised into direct healthcare resources that include medical expenditures and direct non-healthcare resources like transportation, household expenditures and informal care[6].
Indirect costs	The costs incurred due to a loss of productive output, or 'productivity loss', due to disability or premature death [6 59].
Human capital method	Considers the patients hours of productivity that are lost and designed to estimate the value of human capital by calculating productivity costs as a product of those total lost hours with the hourly wage[6 57].
Friction cost method	Estimates the value of human capital when another athlete or worker from the employment pool replaces the present value of a workers future earnings until the injured worker returns or is replaced[6]. This only counts those hours until another employee takes over the patients work, or the 'friction period' and depends on the availability of labour[6 57].
Top-down approach	Use of overall expenditures and then allocating costs using formulae to estimate unit procedure or service costs as a proportion of the total known expenditure[52 56].
Bottom-up approach	Uses data from records at a service provider level to estimate unit costs, whereby total costs come out through the multiplication of unit costs by the quantities used[6 52].
Econometric approach	Estimates the difference in costs between an injured cohort and a non-injured cohort, whereby the two cohorts are matched usually through a series of regression analysis[6].
Cost perspective	Determined from a variety of bearers of healthcare costs[7]. These perspectives may be from a cost to a society, business, or business sector, third party payers, hospital, healthcare system or the patient and their families[5-7]. Each perspective offers information about the costs to a particular group[6].

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Table 2

Characteristics of included studies

Study	Study design	Sport	Injury area	Type of cost	Costing approach	Cost perspective
Sabesan et al[15] 2022	Retrospective	Major League Baseball (MLB)	Shoulder	Indirect	Top-down	Sports organisation
Yakal et al[38] 2021	Prospective	Not specified	Multiple	Direct	Bottom-up	Healthcare system, hospital
Sleeswijk et al[41] 2021	Prospective	Running	Multiple	Direct and indirect	Bottom-up	Societal
Lu et al[42] 2021	Prospective	Football (A-League)	Not specified	Indirect	Top-down	Sports organisation
Vaudreuil et al[34] 2021	Retrospective	Basketball (NBA)	Knee (ACL)	Indirect	Top-down	Sports organisation
Lam et al[36] 2021	Retrospective	Multisport	Knee	Direct	Bottom-up	Athlete or Athlete's families
Meldau et al[21] 2020	Retrospective	Major League Baseball (MLB)	Elbow	Indirect	Top-down	Sports organisation
Eliakim et al[23] 2020	Retrospective	Football (EPL)	Multiple	Indirect	Top-down	Sports organisation
Ackerman et al[26] 2020	Retrospective	Not specified	Hip	Direct	Bottom-up	Healthcare system, hospital
Marshall et al[27] 2020	Retrospective	Multisport	Ankle	Direct	Bottom-up	Athlete or athlete's families
Gebert et al[29] 2020	Retrospective	Football (soccer)	Multiple	Direct and indirect	Bottom-up	Third-party payer
Robinson et al[31] 2020	Retrospective	Multisport	Wrist/ Hand	Direct	Bottom-up	Healthcare system, hospital
King et al[28] 2019	Retrospective	Rugby Union	Multiple	Direct and indirect	Bottom-up	Third-party payer
Lystad et al[32] 2019	Retrospective	Not specified	Multiple	Direct	Bottom-up	Hospital
Adler et al[37] 2019	Retrospective	Horse racing; Other	Not specified	Direct	Bottom-up	Hospital
Navarro et al[16] 2018	Retrospective	Ice Hockey (NHL)	Head (concussion)	Indirect	Econometric	Sports organisation

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Table 2 (*continued)*

Study	Study design	Sport	Injury area	Type of cost	Costing approach	Cost perspective
Navarro et al[17] 2017	Retrospective	Gridiron football (NFL)	Head (concussion)	Indirect	Econometric	Sports organisation
Hespanhol et al[40] 2017	Prospective	Running	Multiple	Direct and indirect	Bottom-up	Societal
Tranaeus et al[43] 2017	Prospective	Floorball	Multiple	Direct and indirect*	Bottom-up	Societal
Hiploylee et al[18] 2016	Retrospective	Ice Hockey (NHL)	Head (concussion)	Indirect	Top-down	Sports organisation
Roe et al[20] 2016	Retrospective	Gaelic games	Multiple	Direct and indirect	Bottom-up	Third-party payer
Hespanhol et al[44] 2016	Prospective	Running	Not specified	Direct and indirect	Bottom-up	Societal
Hespanhol et al[39] 2016	Prospective	Running	Not specified	Direct and indirect	Bottom-up	Societal
Finch et al[25] 2015	Retrospective	Not specified	Multiple	Direct	Bottom-up	Healthcare system, hospital
Brown et al[45] 2015	Prospective	Rugby Union	Multiple	Direct	Bottom-up	Athlete or Athletes families
Donaldson et al[24] 2014	Retrospective	Ice Hockey (NHL)	Multiple	Indirect	Top-down	Sports organisation
Hickey et al[30] 2014	Retrospective	Australian Football (AFL)	Thigh (hamstring)	Indirect	Top-down	Sports organisation
King et al[33] 2011	Retrospective	Rugby League	Multiple	Direct and Indirect	Bottom-up	Third-party payer
Cumps et al[19] 2008	Retrospective	Multisport	Multiple	Direct and indirect	Bottom-up	Societal
Turner et al[22] 2008	Retrospective	Horse racing	Multiple	Direct and indirect	Top-down	Third-party payer
Forssblad et al[35] 2005	Retrospective	Multisport	Knee	Direct	Bottom-up	Third-party payer

*Tranaeus et al[43] stated their aim to collect direct and indirect costs, however noted that due to too few records it was not possible to include sick leave and productivity costs

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Table 3 Types of costs and resources

Direct healthcare costs	Physician services[19 20 22 26 28 29 39-41 43-45]
	Ancillary services[19 20 26 28 29 39-41 43-45]
	Athletic Trainers[27 36]
	Hospital inpatient services[19 20 25 26 31 32 35 37 38 45]
	Rate and duration of hospital stay[31 32 37]
	Hospital outpatient services[19 20 25 26 28 31 32 37 38]
	Diagnostic investigation[19 20 28 37 43 45]
Direct non-healthcare costs	Transportation costs[28]
Indirect costs	Missed matches[15-18 21 24 30 34 42]
	Work absenteeism[19 20 22 28 29 39-41 44]
	Team worth[23]

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CRediT Author Statement

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Declaration of interests

The authors of this review declare that there are no competing interests. The authors also declare that this scoping review is not under consideration for publication elsewhere, that its publication is approved by all authors and that if accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright-holder.

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Confirmation of ethical compliance

This review did not require human participants.

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